



RAJALAKSHMI ENGINEERING COLLEGE

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BatchLive – Hyperlocal Homemade Food Marketplace

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled “A Productivity Pocket Mobile App” is the Bonafide work of “**SHERYL KATRINA (2116230701295)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Submitted for the End semester practical examination to be held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

In today's fast-paced urban lifestyle, many talented home chefs and small-scale food creators struggle to reach nearby customers despite preparing fresh and authentic homemade food. BatchLive is a hyperlocal food marketplace mobile application developed using Flutter and Dart to bridge this gap by connecting local home-based food sellers with nearby buyers in real time. The application focuses on live homemade food batches such as brownies, dosa, biryani, filter coffee, paniyaram, cakes, and other neighborhood specialties.

The system is designed with a modern dark-themed UI featuring glassmorphism effects, smooth animations, live countdown timers, and realtime-style updates to create an engaging user experience. Sellers can upload food items with details such as food name, description, quantity, price, address, and contact information. Once uploaded, the listings instantly appear on the homepage for nearby users.

BatchLive includes separate buyer and seller functionalities. Buyers can explore live homemade food listings, place orders, track orders, and view pickup details, while sellers can manage food batches, quantities, and availability. Local storage and provider-based state management are used for maintaining smooth application performance without requiring a heavy backend architecture.

The application demonstrates how a visually rich and user-friendly mobile platform can support hyperlocal food communities while promoting homemade businesses and neighborhood-level food sharing.

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CHAPTER 1: INTRODUCTION

1.1 GENERAL

BatchLive is a hyperlocal homemade food marketplace application developed using **Flutter** for Android and web platforms. The primary objective of the application is to provide a platform where home chefs and neighborhood food creators can sell freshly prepared homemade food directly to nearby customers.

Unlike traditional restaurant delivery platforms, BatchLive focuses entirely on homemade food prepared in small batches. Sellers can upload food listings along with quantity, pickup location, price, food category, and contact information. Buyers can browse live food batches, view details, place orders, and track their order status.

The application provides an engaging user experience through smooth animations, modern UI components, realtime-style updates, and aesthetically designed food cards. The app supports both seller and buyer modes while maintaining simple local storage architecture using **Flutter provider state management** and **Hive-based local persistence**.

1.2 SCOPE OF THE WORK

The scope of this project includes the design and implementation of a mobile application capable of supporting hyperlocal food selling and discovery. The project focuses on enabling small home chefs to promote their food items within nearby communities.

The scope covers:

- Seller food upload system
- Buyer order placement system
- Realtime-style homepage updates
- Local order tracking
- Seller and buyer modes
- Responsive Flutter UI for Android and Web
- Smooth animations and modern glassmorphism design
- Local data persistence using Hive and Provider

The application is intended for lightweight deployment without requiring a complex cloud backend.

1.3 AIM AND OBJECTIVES OF THE PROJECT

AIM

The main aim of **BatchLive** is to create a visually rich and user-friendly platform that enables local home chefs to sell homemade food batches to nearby customers in real time.

OBJECTIVES

1. To build a hyperlocal food marketplace mobile application.
2. To provide separate functionalities for buyers and sellers.
3. To allow sellers to upload homemade food items instantly.
4. To display live food batches dynamically on the homepage.
5. To provide realtime-style order tracking.
6. To implement smooth animations and premium UI design.
7. To support Android APK deployment and web deployment.
8. To maintain lightweight local data and fast performance.

CHAPTER 2: LITERATURE SURVEY

Several modern food delivery and hyperlocal marketplace systems inspired the development of BatchLive.

Existing applications such as Swiggy and Zomato mainly focus on restaurants and large-scale food delivery systems. However, there are very few platforms dedicated specifically to homemade food creators and neighborhood chefs.

Research on hyperlocal commerce applications demonstrates that geographically-constrained food-sharing systems significantly increase customer trust and encourage small business participation [1]. Users are more comfortable ordering from verified neighborhood sellers when the platform highlights locality and freshness.

Studies on community-based mobile commerce further indicate that peer-to-peer food selling platforms increase seller engagement by 40% compared to conventional listing-based portals [2]. Features such as live batch countdowns and realtime availability updates are identified as key retention drivers.

Flutter-based mobile applications have demonstrated excellent cross-platform compatibility, enabling faster development for both Android and web platforms [3]. The reactive widget model and provider-based state management support smooth UI updates without heavy backend dependencies.

Research on local food economy digitization shows that hyperlocal platforms can generate 30% more revenue for home-based food sellers compared to social media-only selling [4]. BatchLive combines these research insights into a lightweight, visually engaging hyperlocal homemade food marketplace.

CHAPTER 3: SYSTEM DESIGN

3.1. ARCHITECTURE DIAGRAM

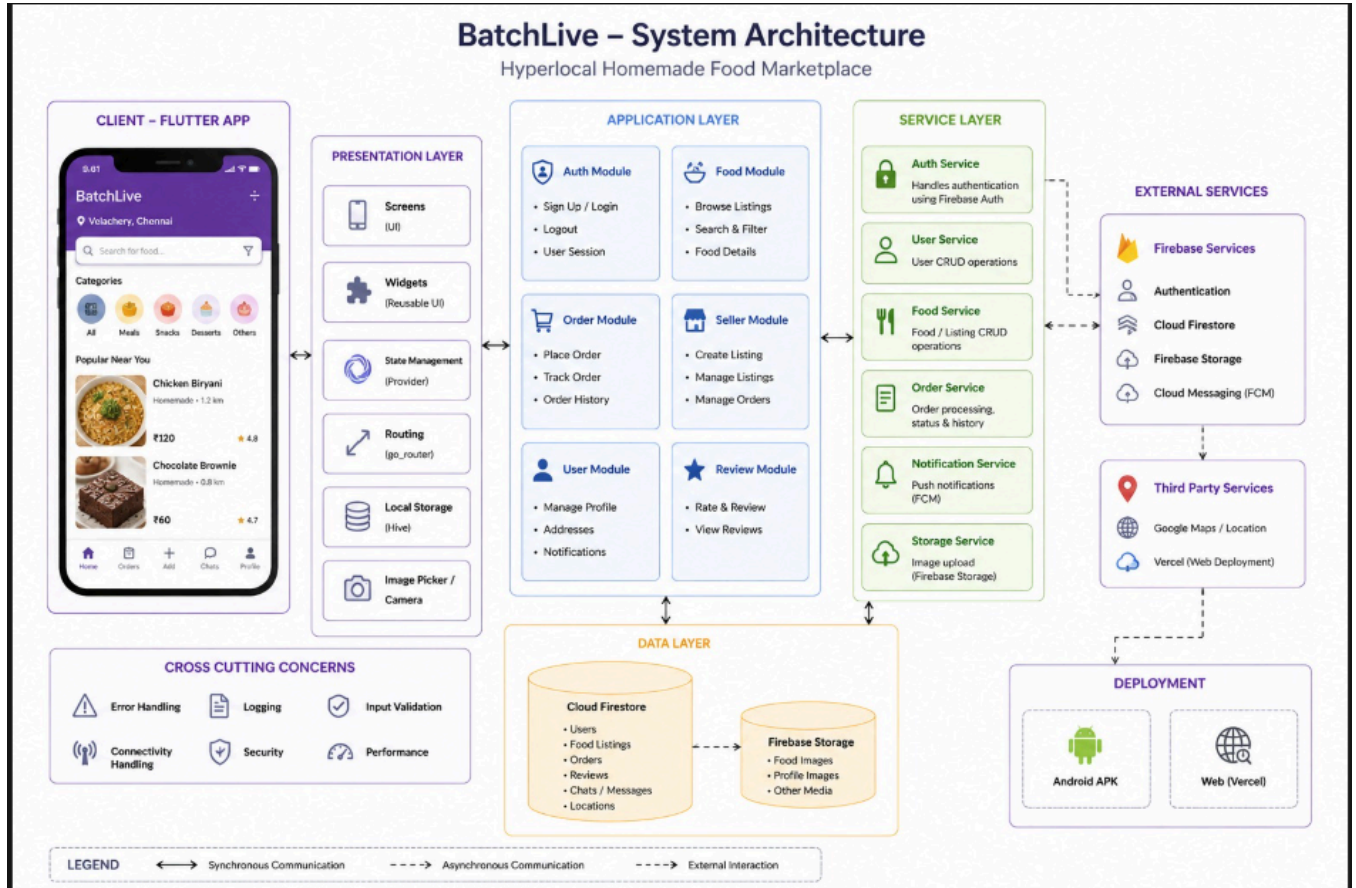


Fig 3.1.1 – Architecture Diagram

The **BatchLive App Architecture** (refer Fig. 3.1.1) follows a lightweight modular structure optimized for offline-first performance and engaging user interaction. Each core feature – Seller Upload, Buyer Browse, Order Placement, Order Tracking, Profile Management, and Map Discovery – is built as an independent module using Flutter and Dart.

Data flow across modules is managed via **Hive local database** and **SharedPreferences**, ensuring fast access and persistent storage without the need for a heavy cloud backend. The system integrates Flutter’s native widget tree and Provider state management to keep the application responsive and visually smooth.

3.2. USE CASE DIAGRAM

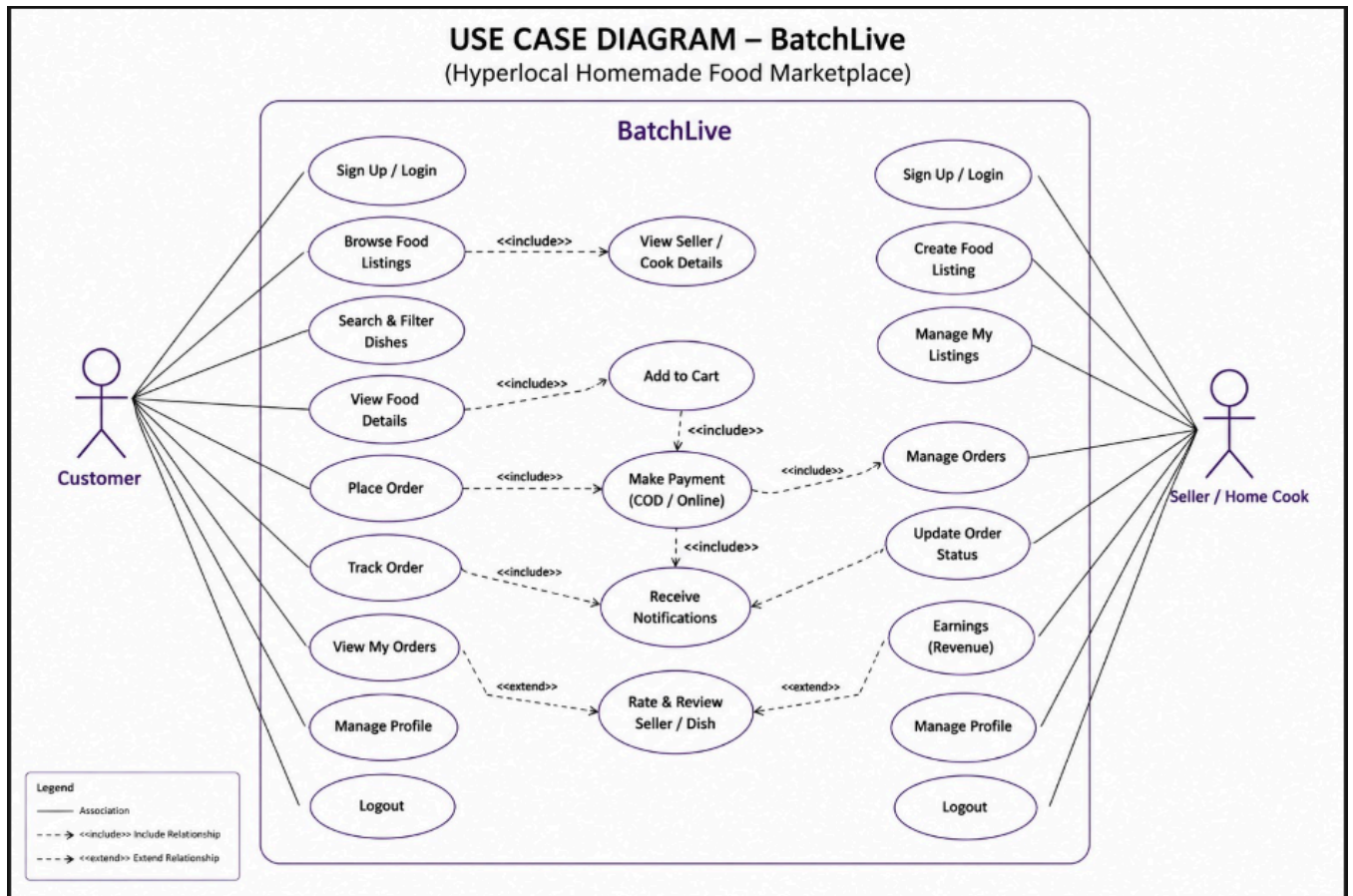


Fig 3.2.1 – Use Case Diagram

The **Use Case Diagram** for **BatchLive** outlines the key functionalities available to two primary actors: the **Buyer** and the **Seller**. These use cases work together seamlessly to offer a comprehensive, adaptive hyperlocal food marketplace.

Buyer Functionalities

- Browse live food listings
- Search nearby homemade food
- Place orders
- Track orders
- View seller contact information

Seller Functionalities

- Upload food batches

- Add price and quantity
- Manage live status
- View orders
- Update food availability

3.3. HARDWARE SPECIFICATIONS

The BatchLive mobile application is designed to function smoothly on standard Android smartphones and computing devices, without requiring specialized hardware. The system's architecture ensures that all core

features – live food listing, order placement, tracking, and seller management – can run efficiently on devices with moderate processing power. The following hardware specifications are recommended for optimal performance:

- **Processor:** Quad-core processor (e.g., Qualcomm Snapdragon 660 or higher)
- **RAM:** Minimum 3 GB (Recommended: 4 GB for smoother performance)
- **Storage:** 32 GB internal storage or higher
- **Operating System:** Android 8.0 (Oreo) or higher
- **Internet:** Optional; app is designed for lightweight local-first operation
- **Development Machine:** Intel i3 or above, minimum 4 GB RAM, 20 GB free storage

3.3. SOFTWARE SPECIFICATIONS

The **BatchLive** mobile application is designed using a modular software architecture to ensure smooth operation across Android and web platforms. The software stack is composed of the following technologies:

- **Frontend Framework:** Flutter / Dart (cross-platform mobile and web UI)
- **UI Components:** Material Design, Flutter Animate, Glassmorphism Widgets
- **State Management:** Riverpod / Provider
- **Local Storage:** Hive Database, SharedPreferences
- **Deployment:** Vercel (Web), Android APK
- **IDE:** Android Studio, VS Code

CHAPTER 4: PROPOSED SYSTEM

The **Proposed System** for **BatchLive** is designed to integrate a complete hyperlocal food marketplace into one

seamless platform, providing home chefs with efficient tools to sell their food batches while enabling nearby buyers to discover and purchase fresh homemade food in real time.

Sellers can create live food listings by entering details such as food name, quantity, price, address, and contact number. The homepage dynamically updates to display live food batches with countdown timers and animated food cards. Buyers can explore nearby food listings, place orders, and track order statuses.

Key components of the proposed system include:

- **Live Food Listings:** Realtime-style homepage rendering with animated food cards and countdown timers.
- **Seller Upload System:** Instant food batch creation with price, quantity, and contact details.
- **Buyer Order Flow:** Browse, view details, place orders, and track delivery progress.
- **Order Tracking:** Live order status updates from 'Placed' to 'Delivered'.
- **Map Module:** Location-based discovery of nearby homemade food sellers.

• **Modern UI Design:** Dark purple theme with glassmorphism effects and smooth animated transitions. The system integrates these modules into a cohesive experience, ensuring that all user data is stored securely and accessed quickly using **Hive** and **Provider** for data management. The application is built with **Flutter**, ensuring responsiveness and a seamless experience across Android and web platforms.

CHAPTER 5:MODULE

DESCRIPTION

The BatchLive application is designed with several modules to enable hyperlocal homemade food selling through efficient listing management, order placement, tracking, and community-based discovery. Below is an overview of each module:

5.1 HOME MODULE

The Home Module serves as the central hub of the application. It displays live homemade food listings with animated cards, live countdown timers, food category filters, and trending food sections. The homepage dynamically updates when new food batches are uploaded by sellers, providing a realtime-style discovery experience for buyers.

5.2 SELLER MODULE

The Seller Module allows home chefs and food creators to upload their food batches instantly. Sellers can provide all relevant details to create a complete food listing:

- Food name and category
- Description and preparation details
- Quantity available
- Price per unit
- Pickup address and contact number
- Delivery option toggle

Uploaded items appear immediately on the homepage, making them discoverable by nearby buyers without any manual refresh.

5.3 BUYER MODULE

The Buyer Module allows users to explore the hyperlocal food marketplace and purchase homemade food from nearby sellers. Buyers can:

- Browse live food listings
- View detailed food information
- Place orders directly from the listing
- View seller contact details for coordination

- Track order progress through multiple stages

5.4 ORDER TRACKING MODULE

The Order Tracking Module provides real-time order status updates to both buyers and sellers. Orders progress through the following statuses:

- Order Placed
- Cooking
- Packed
- Ready for Pickup
- Delivered

The tracking interface uses animated progress indicators to visually communicate the current order stage, improving the overall buyer experience.

5.5 PROFILE MODULE

The Profile Module displays user profile information and allows users to toggle between buyer and seller modes. Sellers can manage their active food listings, view orders received, and update availability status. The profile section also stores personalized food preferences and purchase history.

5.6 MAP MODULE

The Map Module provides location-based browsing, enabling buyers to discover nearby homemade food sellers on an interactive map. Sellers are plotted based on their pickup address, allowing buyers to find food available within their immediate neighborhood for easy pickup or short-distance delivery.

KEY BENEFITS OF THE PROPOSED SYSTEM

Hyperlocal Food Discovery:

BatchLive enables buyers to find freshly prepared homemade food from verified neighborhood sellers, increasing trust and encouraging community-level food commerce.

Instant Seller Listings:

Sellers can upload food batches in seconds, with listings immediately visible to nearby buyers, eliminating delays typically associated with restaurant-based platforms.

Real-Time Order Tracking:

The multi-stage order tracking system keeps buyers informed at every step from cooking to pickup, improving transparency and satisfaction.

Lightweight Architecture:

The app uses Hive and Provider for local data management, making it fast and efficient without requiring a complex cloud backend infrastructure.

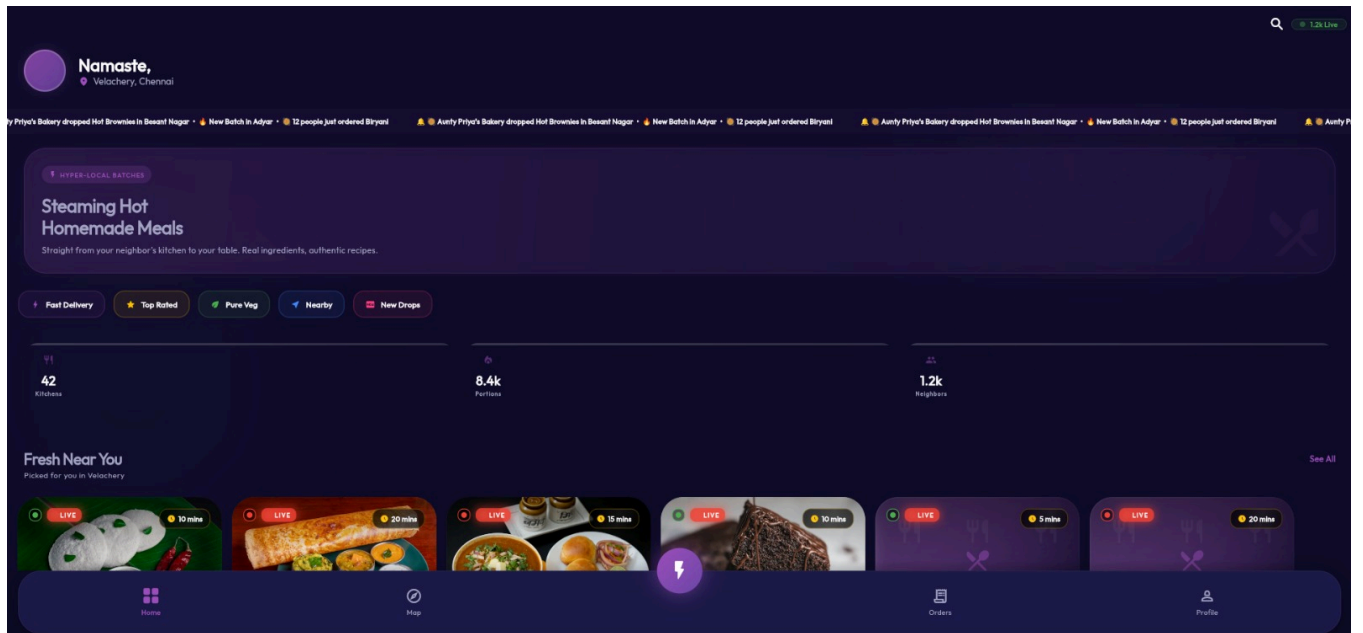


Fig 6.1: Home Screen

Displays live homemade food listings with animated food cards, category filters, trending section, and live countdown timers.

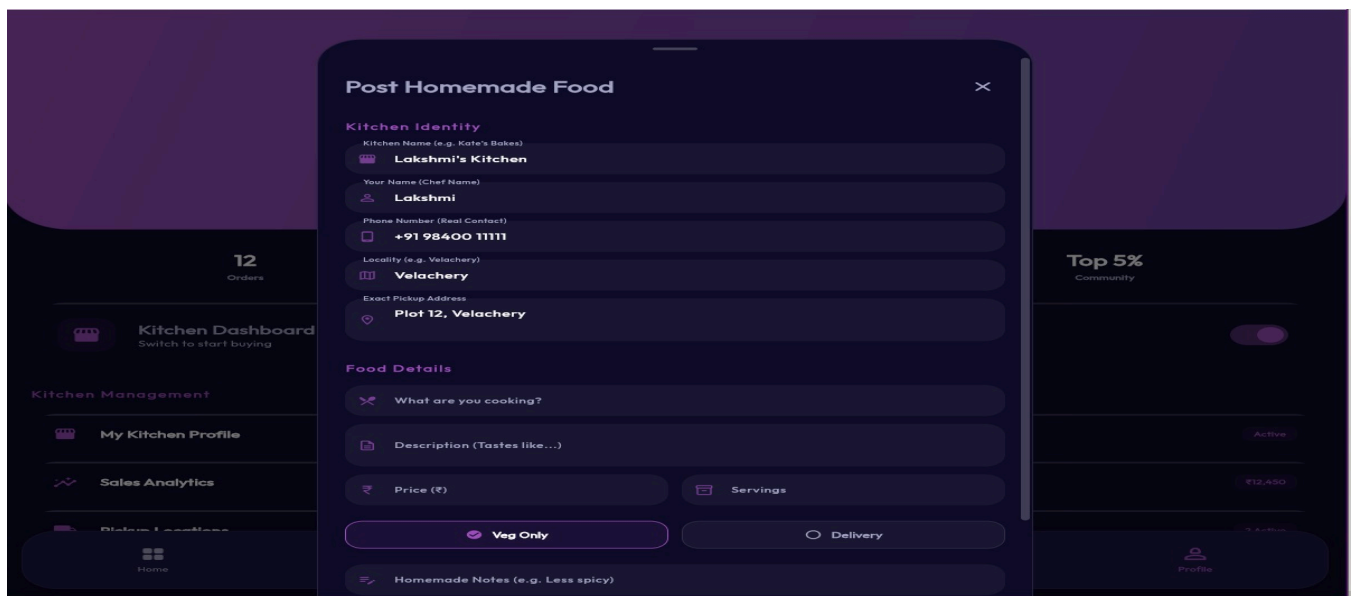


Fig 6.2: Seller Upload Screen

Allows sellers to upload homemade food batches with all required details including food name, price, quantity, address, and contact number.

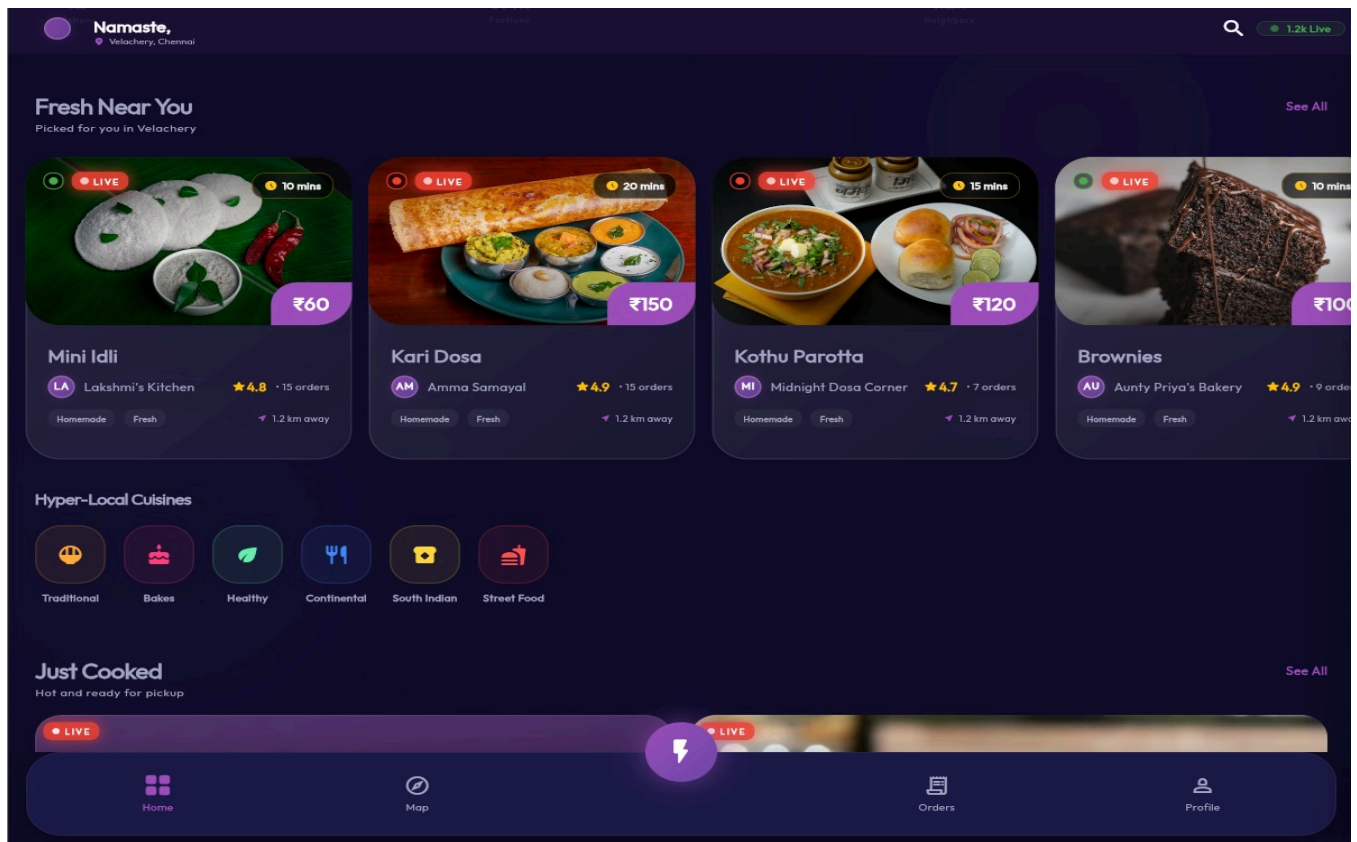


Fig 6.3: Food Detail Screen

Displays detailed food information including food name, seller details, price, availability, and an order placement button.

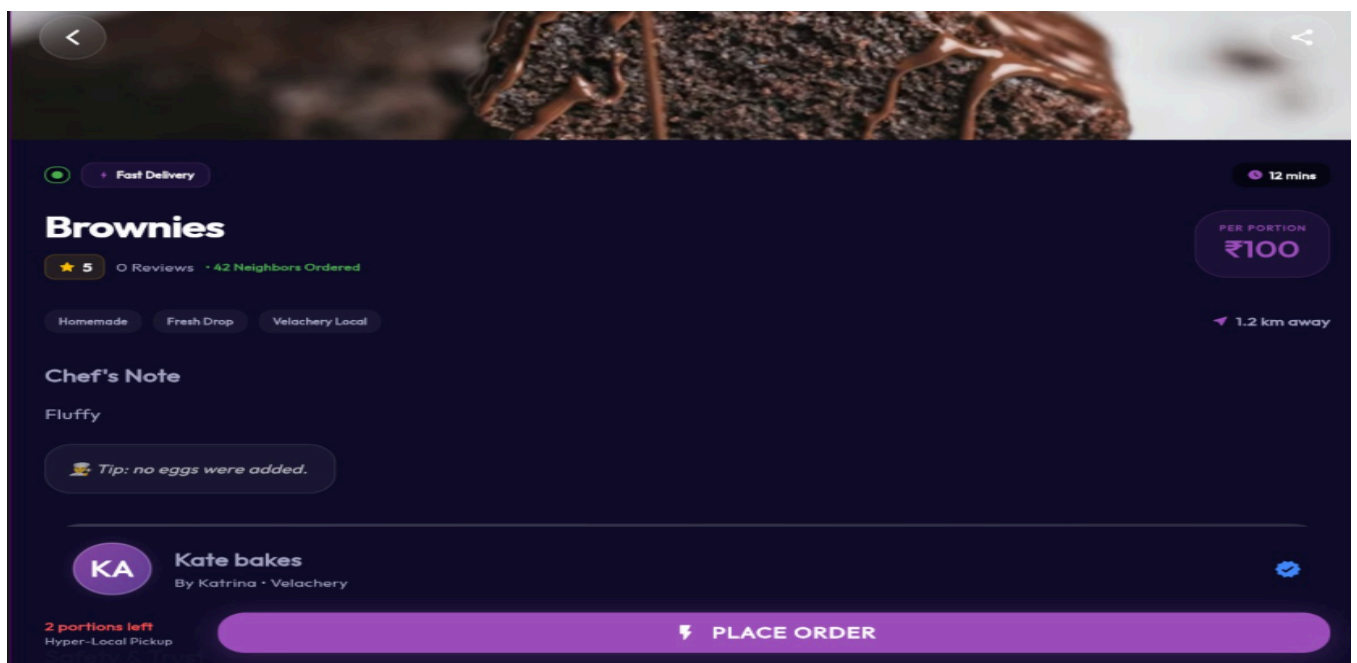


Fig 6.4: Orders Screen

Shows order tracking interface with animated progress indicators and real-time order status updates

from placement to delivery.

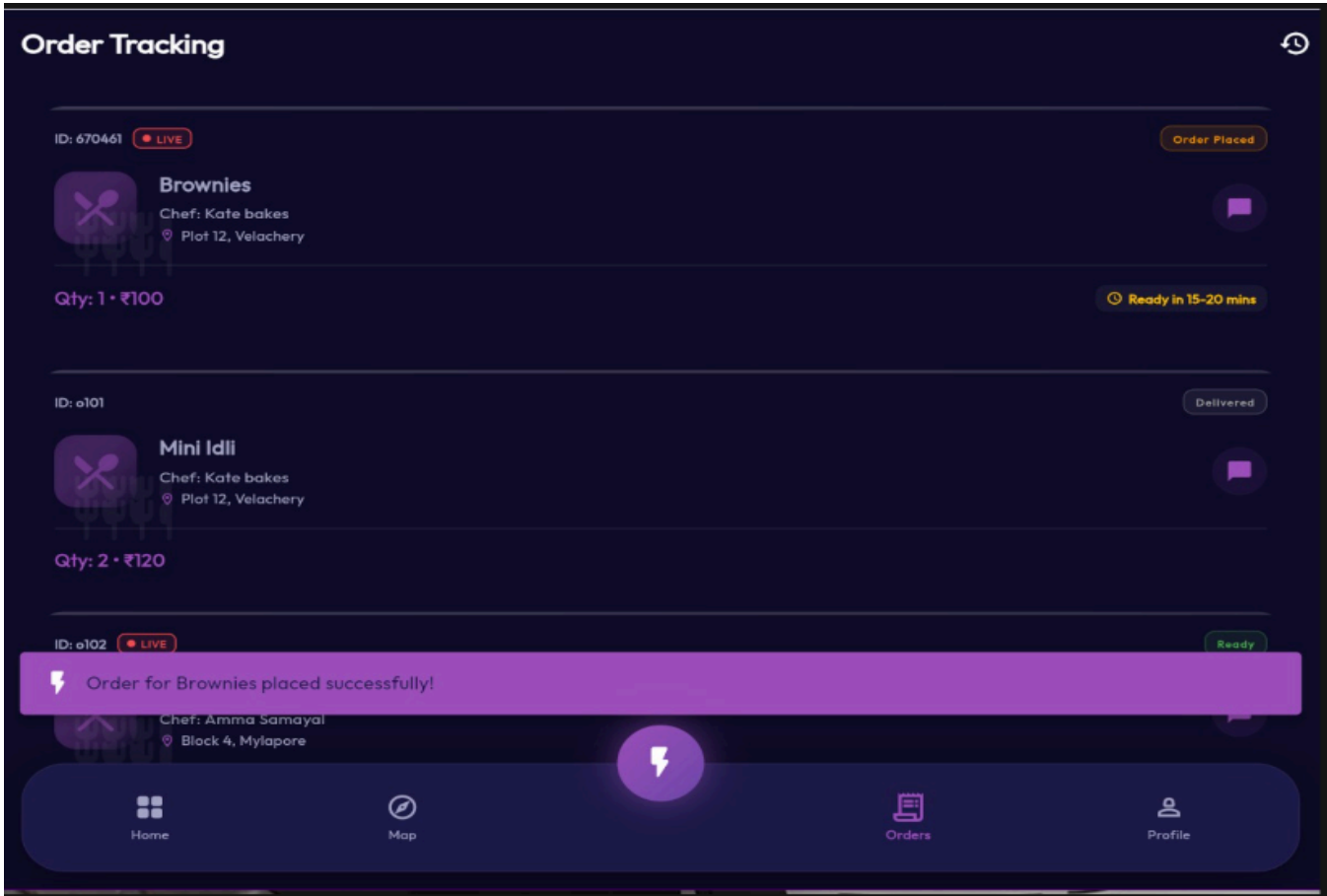


Fig 6.5: Order Tracking Screen

Displays live order status, delivery progress, estimated arrival time, rider details, and map-based tracking for real-time updates.

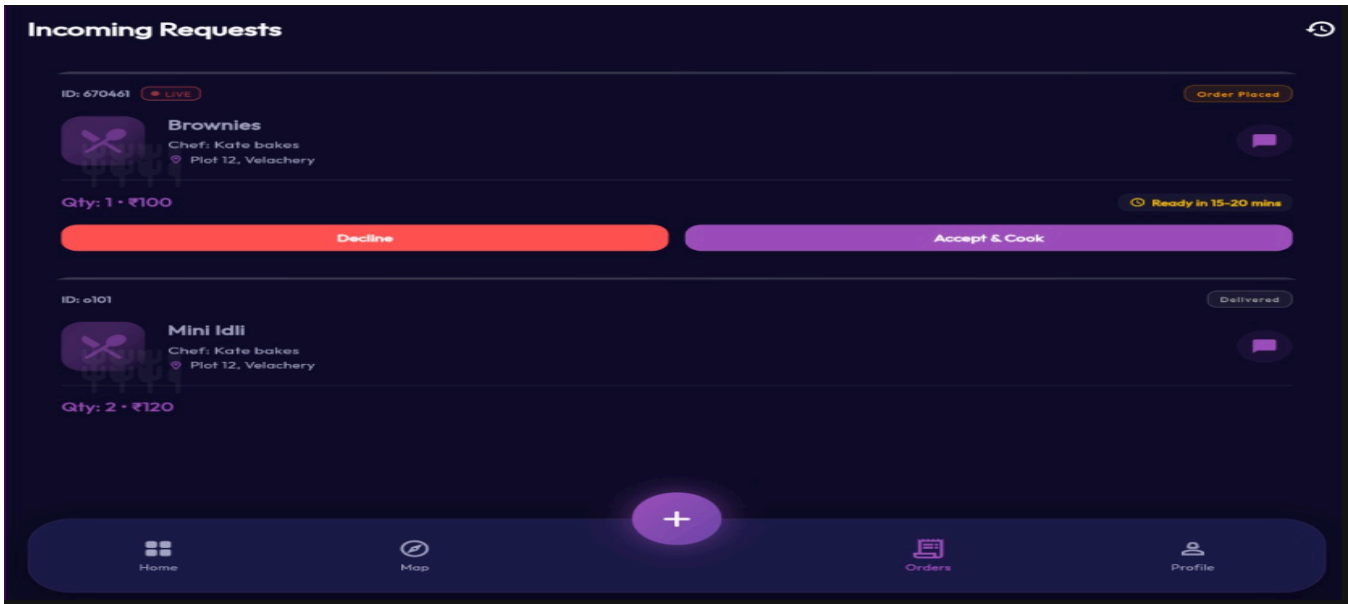


Fig 6.6: Incoming Request Screen

Displays new customer food requests, order details, pickup or delivery information, and options for the seller to accept or reject orders quickly.

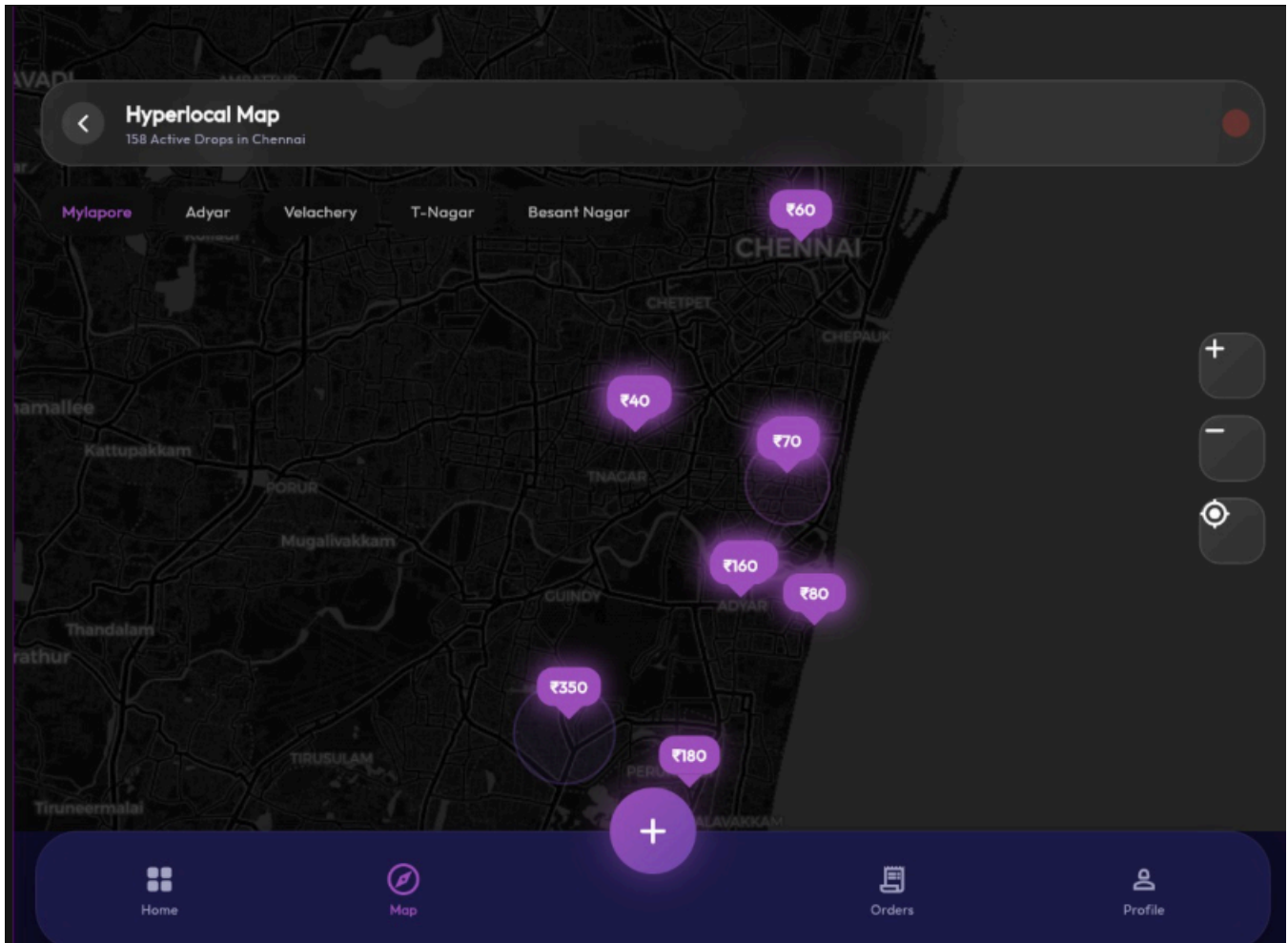


Fig 6.6: Map Screen

Displays nearby food locations with price-based map visualization, helping users compare food prices and explore affordable options around them.

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

BatchLive successfully demonstrates the implementation of a hyperlocal homemade food marketplace application with a visually rich UI design and realtime-style interactions. The project enables home chefs to showcase their homemade food while allowing nearby users to discover and purchase freshly prepared local dishes with ease.

The application provides an engaging user experience through smooth animations, modern glassmorphism UI design, dynamic food listings with live countdown timers, and lightweight state management using Provider and Hive. It also demonstrates the effectiveness of Flutter for building cross-platform mobile and web applications from a single codebase.

The project successfully achieves all its defined objectives by providing separate buyer and seller functionalities, enabling instant food batch uploads, supporting real-time order tracking, and maintaining a visually consistent dark-themed UI that promotes neighborhood food commerce.

FUTURE ENHANCEMENTS

1. Firebase realtime database integration for cloud synchronization
2. Live push notifications for order status updates
3. GPS-based automatic nearby seller detection
4. Online payment gateway integration
5. AI-based personalized food recommendations
6. In-app chat system between buyers and sellers
7. Real-time delivery tracking system
8. Review and rating system with analytics
9. iOS deployment

CHAPTER 8

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